

ASSESSMENT GUIDE / MARKING SCHEME

Specialist Mathematics Units 1/2 - Kinematics: Felix's Party Quiz Showdown

Award marks for mathematically equivalent working. Consequential working may be credited where appropriate. Exact values are required unless otherwise indicated.

Question 1 - 4 marks

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|---|----------|
| a. Differentiates correctly: $v(t) = 3t^2 - 18t + 24$ and $a(t) = 6t - 18$. | 1 |
| b. Sets $v(t) = 0$ and factorises $3(t-2)(t-4)=0$. | 1 |
| Obtains $t = 2$ s and $t = 4$ s. | 1 |
| c. Factorises $x(t)=(t-1)(t-4)^2$, so the first centre crossing is $t=1$; hence $a(1)=-12$ m/s ² . | 1 |

Question 2 - 5 marks

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|--|----------|
| a. Integrates acceleration: $v(t) = 9t^2 - 24t + C$. | 1 |
| Uses $v(1)=-2$ to obtain $C=13$, hence $v(t)=9t^2-24t+13$. | 1 |
| b. Finds the velocity zeros in $[0,2]$: $t = (4 \pm \sqrt{3})/3$. | 1 |
| Uses an antiderivative of velocity, $x(t) = 3t^3-12t^2+13t+C$, and splits the interval at both turning times. | 1 |
| Adds the absolute displacements over the three intervals to obtain total distance $2 + 8\sqrt{3}/3$ m. | 1 |

Question 3 - 7 marks

a. Finds displacement from 0 to 4: $\int_0^4 \frac{1}{2}t^2 dt = 32/3$. 1

Finds displacement from 4 to 7: $\int_4^7 (24-4t) dt = 6$. 1

Combines the displacements correctly: total displacement = 50/3. 1

Hence obtains $x(7) = 100 + 50/3 = 350/3$ m under the stated signed-position convention. 1

b. Shows $v(6)=0$ and that v changes sign from positive for $4 < t < 6$ to negative for $6 < t \leq 8$. 1

Finds distances $0-4 = 32/3$, $4-6 = 8$, and $6-8 = 8$. 1

Total distance = $32/3 + 8 + 8 = 80/3$ m. 1

Question 4 - 5 marks

a. Equates velocities: $4t-4 = 3t^2$, giving $3t^2-4t+4=0$. 1

Notes the discriminant is $16-48=-32 < 0$, so there is no real time $t > 0$ at which the velocities are equal. 1

b. Finds $a_M=4$ and $a_K=6t$; solves $6t=3(4)$ to obtain $t=2$. 1

Evaluates positions: $x_M(2)=12$ and $x_K(2)=12$. 1

Distance between the players = 0 m. 1

Question 5 - 4 marks

a. Integrates velocity to express the displacement from 0 to 5: $\int_0^5 k(t-5)^2 dt = 125k/3$. **1**

Uses displacement -50 m to obtain $125k/3 = -50$, hence $k = -6/5$. **1**

b. Obtains height $x(t) = -2/5(t-5)^3$, and solves $x=25$ to get $t-5 = -5/\sqrt[3]{2}$. **1**

Differentiates velocity: $a(t) = 2k(t-5)$, giving $a = 12/\sqrt[3]{2} = 6\sqrt[3]{4} \text{ m/s}^2$. **1**

TOTAL: 25 MARKS